SAMPLE

Cosmos Unveiled: Supernovas

When Stars Explode

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Introduction

Every few seconds, somewhere in the observable universe, a star ends its life in the most violent event nature permits inside a galaxy: a detonation that, for a few weeks, outshines the combined light of everything else around it, a single star visible across billions of light-years, a factory for the elements, a clock for the cosmos, and a death that seeds the chemistry of the next generation of worlds. This book is about that event, the supernova: what it is, what makes it, what it makes, how we found them, and what they have taught us.

The supernova is astronomy's great confluence. It is a stellar event, the end of a star's story, and anyone who wants the whole stellar biography will find it in the companion volumes on the stars themselves. It is a chemical event, the origin of most of the periodic table's heavy half, the iron in the planet's core and the oxygen in the visitor's blood forged in these fires. It is a cosmological event, the standard candle whose brightness revealed the universe's acceleration, the discovery that remade physics in the visitor's lifetime. And it is a local event, the explosions whose debris is in the ocean floor and the local bubble of gas the Sun travels through, the history written in our own atoms and our own sky.

The book's chapters run from the oldest records to the newest questions. The visiting stars of the historical records, the guest stars the old astronomers logged for centuries without knowing what they were. The classification, the field's great sorting of the explosions into types. The two routes to destruction, the thermonuclear runaway of the white dwarf and the core collapse of the massive star, each with its own physics and its own products. The neutrino flash, the one modern supernova caught at close range. The nucleosynthesis, the elements' forge. The standard candles and the accelerating universe. The surveys, from the lone astronomers of the photographic era to the robots that find thousands a year. The progenitors, the stars we have caught in the act. The neighborhood, the candidates and the debris. The danger, honestly assessed. And the questions the field still cannot answer, which are many, and which are the reason the next decade's machines are being built.

The newcomer should bring one habit to this book,

A word about the book's neighbors, because the subject has edges this volume respects. The star's life before the explosion, the full biography of the burning, is the companion volume's subject, and the leftovers after the explosion, the expanding clouds and the compact bodies, have their own treatments on the shelf, and this book holds the explosion itself: the event, the mechanisms, the light, the uses, and the history, with the before and the after pointed to and given their own pages elsewhere, the newcomer who collects the set owns the whole life, the death, and the estate. The field's own habit: everything here was observed somewhere, by someone, with an instrument, and the field's history is a story of instruments as much as explosions, the telescopes and the spectrographs and the neutrino tanks and the satellites that caught what the eye could never see. The supernova was a mystery for a thousand years because the records were better than the theory, and it has been a science for a century because the theory finally had the tools. The chapters that follow are that story.

Chapter 01: The Visiting Stars: A Thousand Years of Records

The supernova entered history as a problem in the archives, a new star that appeared, stayed, and vanished, and this chapter is the record of those appearances: the guest stars of the Chinese, Japanese, Korean, Arabic, and European observers, and the slow recognition of what they meant.

The records are older than the understanding by a millennium, and the tourist should meet them as the observers left them. The first supernova the historians accept from the records is the star of the year 185, logged by the Chinese astronomers as a new star in the southern sky, a guest star that stayed eight months, the event whose remnant the modern telescopes have found, the oldest recorded explosion. The great one of 1006, the brightest star in the historical record, seen across the hemisphere, described by the Egyptian physician as a spectacle of the sky, the object whose distance the modern instruments have measured at seven thousand light-years, the brightest thing humans have ever seen in the night besides the Sun. The star of 1054, the guest star the Chinese logged for two years,

The 1054 event takes a paragraph of its own, because it is the records' richest case. The guest star was logged by the Chinese astronomers and, the field's historians found, by the Japanese and possibly in an Arab chronicle and an Ancestral Puebloan petroglyph's candidate, the star visible in daylight for weeks and at night for two years, the records' most persistent visitor, and the modern identification, made in the twentieth century, connected the records to the object the telescopes found in the same part of the sky: the expanding cloud, the tangled remnant

whose center holds the spinning star, the whole system the field has studied for a century as the guest star's corpse, the event whose identity the historians and the astronomers established together, the oldest supernova whose remnant we can watch expanding today, the records and the physics holding hands across nine hundred years. The event whose expanding cloud became the sky's most studied object, the pulsar's home. And the two that broke the European worldview: the star of 1572, which the young Danish astronomer measured night after night with the finest instruments of the age and showed, by its lack of parallax, to be beyond the Moon's sphere, in the supposedly unchanging heavens, and the star of 1604, which the imperial mathematician in Prague studied in the last year before the telescope, the star that appeared in the constellation of the serpent holder and was gone by the next autumn.

The significance of 1572 gets the chapter's centerpiece,

The parallax method merits a paragraph to itself, because the argument the young observer made was the era's instrument. The new star's position was measured against the fixed stars night after night, and the demonstration's logic a visitor can run: any object near the Earth, within the sphere of the Moon's region, would show the daily parallax, the shift the observer's own motion across the sky produces, and the star showed none, its position steady against the distant background to the limit of the era's instruments, the star too far for the sublunar realm, and the newcomer can keep the argument's place in the century's history: the measurement that placed the changing object in the heavens, the star that moved the boundary of the mutable, the parallax's absence the presence of the revolution, the supernova as the astronomical method's first great demonstration. Because it is the first supernova of the scientific era and the last of the pre-telescopic one. The young observer, Tycho by name, published his analysis, the measurements that placed the new star beyond the Moon's region, in the sky's eighth sphere, and the conclusion, printed for the age to read, was the demolishing one: the heavens change, the celestial regions are not the incorruptible perfection the schools taught, the star that appears and vanishes proves the sky itself is subject to generation and decay, and anyone should sense the era's shock honestly, because this was one of the cracks the old worldview broke along, the supernova as a founding instrument of the scientific revolution, an explosion in the sky that helped explode a cosmology on the ground.

The gap that followed receives its note,

The frequency's fairness keeps a closing note, because the gap has a statistical defense. The galaxy's rate, by the field's best accounting, is a few supernovae per century, and a four-century gap, a visitor can run the arithmetic, is the coin-flip run the randomness permits: a few per century means a century can pass with none, and four in a row is the unlucky streak the probability allows, the gap's simplest explanation the one that needs no dust, and the field's honest picture holds both: the dust hid some and the luck withheld some, the missing centuries the sum of the screened and the unlucky, the statistics' contribution to the history, the gap partly a coin-flip run, the visitor takes away the lesson's shape, the field's standing caution about the small numbers: the rare events' gaps are not always meanings, sometimes the dice just run quiet.

The hidden explosions earns a paragraph apiece, because the gap's explanation has been tested. The radio and the x-ray surveys, the instruments that see through the dust, have catalogued the remnants of the recent explosions the visible records missed: the young shells in the obscured regions, the candidates' ages in the last few centuries, the explosions that occurred while the telescopes watched other things, and the field's favorite case, the remnant found in the current era whose light arrived, by the models' dating, roughly at the age of the telescopes, the explosion the first observers could have seen had the dust not been in the way, the visitor carries off the gap's honest closure: the four-century silence was partly the galaxy's weather, the dust's screen, and the modern instruments have recovered the missing years' record, the archives' silence filled by the archaeology of the shell. Because it is the field's first great puzzle. After 1604, no supernova appeared in the Milky Way's visible disk for four hundred years, the era of the telescope, the era of the great refractors and reflectors, the era when every other class of object was found, and the brightest phenomenon in the stellar repertoire stayed away, the field's standing irony, and the newcomer can bank the explanation the modern rates imply: the galaxy's disk is partly hidden by dust, and the supernova rate implies several per century, some fraction of which we miss, and the modern instruments, the radio and the x-ray eyes, have found the remnants of the hidden ones, the explosions of the last four centuries that nobody saw, the archive now completed by the telescopes that see through the dust, the missing history recovered.

The chapter closes with the modern debut,

The southern luck carries its note, because the event's audience was the hemisphere's. The explosion's host galaxy rides the southern sky, the companion of the south's folklore, and the event's watchers were the southern

observatories: the Chilean mountains' telescopes, the Australian and the South African stations, the expeditions the northern observatories flew south with their equipment, and the visitor pockets the era's logistics: the field's first instrumented supernova was a southern event, the hemisphere's astronomers the first responders, the whole modern era of the subject opened from the observatories of the southern countries, the debut's geography, a fact the northern reader rarely meets and the book records with pleasure. The supernova of 1987, the first one visible to the naked eye since 1604, the explosion in the southern sky's great companion galaxy, and the event that inaugurated the era this book describes: the supernova observed with every tool the modern field owns, the subject of the chapter where the records and the physics finally meet.

The records' sociology goes on the wall as the shelf's last exhibit. The Chinese and Japanese court astronomers logged the guest stars in official annals; Korean observatories added corroborating entries; the Islamic world recorded a few; and medieval Europe, its learned culture hostile to celestial novelty, logged almost none, a silence the account prices honestly: the sky's most dramatic visitors went unrecorded by the civilization that would later do the most to explain them. The archive the field now reads was written in the court registers of the East, and the chapter's last line credits the clerks: the oldest instrumental era of astronomy was the civil service's bookkeeping.

One record-keeping footnote stays on the wall: the oldest accounts call the visitors guest stars, the term itself a diagnosis, an object where no object was before, and the naming tells the whole story of the pre-telescope eye, which measured position and duration faithfully but owned no instrument capable of the simplest spectra.